

AI Assignment

Bury College RAG Chatbot with Hybrid Q/A Retrieval

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Discovery

Introduction

I'm a Data Science Trainer and AI Agentic trainee. As such I have familiarity with Python for coding, and I also have good understanding of different data structures and the software's used to create both relational databases and data warehouses. I'm also aware of and can use a variety of machine learning techniques which include both supervised and unsupervised learning. This project crosses into data science in terms of data scraping, cleaning of data and the utilising of semantic search and pre-trained generative transformers.

Our objective was to create a chatbot which can answer questions based off data we have scraped off the web. I've chosen Bury College as I am a governor there and if this project is successful, I could expand this out and use it for real. The webpages of this website are publicly available and there is no commercially sensitive information contained within nor are there robots protecting the pages. School and college websites are well known for having too much information on them and by extension being very difficult to navigate for information quickly. A chatbot would help parents, carers, students, staff and governors navigate this website with far more ease.

Constraints

The deadline for this project is 29/05/2026 and it must be submitted on github for review along with this document. I may need to generate an API key, which I have not done before, and I will also be constrained by the model I choose as I want this to be done for free. That will come with drawbacks such as:

- The dimensions the neural network
- The size of the vector database
- The number of times I can send data to the Large Language Model (LLM)
- Where the chatbot will be hosted – this will be a free platform called Streamlit – again, something I have never used before

Success Criteria

I need to select suitable software and then show evidence of the following:

- Use Python (Anaconda JupyterLab)
- Install the relevant libraries and run them into memory
- Data scraping of a website (<https://burycollege.ac.uk/>) and a minimum of 10 webpages
- Clean the data so I'm only left with the text and then save this as .txt or .json file
- Use of a LLM to generate high quality question-answer pairs from the scraped content with a target of between 100 – 200 question and answer pairs
- The Q+A document can be saved as a .csv or .json file
- Create the vector database by chunking the scraped documents
- Create embeddings and then store this in a vector database which will be done in either Chroma or FAISS
- The chatbot itself needs to search the QA database. If the score is below 0.5, then the chatbot will use the Q+A database, if the score is > 0.5 then the system will fall back to all of the pages which have been scraped
- The LLM will then piece together the words it needs for the solutions from the QA database or vector database and return this information back to the user based on the context it has been given
- There needs to be chat history
- The user interface should use Streamlit and feature:
 - chat interface
 - source citation (based on the Q+A document)
 - clear chat button
 - a side bar with project information
 - website link and statistics
 - option to upload new questions or view a sample of the Q+As.

Data Transformation

I will be taking web scrapes from this website: <https://burycollege.ac.uk/>. The scrape will take in the data in HTML format. This poses a problem as I don't want all the HTML tags. This is where beautiful soup comes in useful as it can help me to clean all those tags out. The cleaned data will then be stored as a JSON file (which are good at storing dictionaries with key and value pairings). An example of this is as follows:

```
{ "source_page": "https://burycollege.ac.uk/16-18",  "content": "Bury College 16-18 Courses  
Excellent vocational and academic choices. Applications are now open for September term  
dates entry requirements..." }, etc...
```

These dictionaries are then held in a list for easy retrieval.

Chroma and FAISS Algorithm

This chatbot uses a two-step system to find the best answers. When you ask a question, the system first checks a primary database of 204 pre-made question-and-answer pairs, measuring how close your question is to them using a mathematical "distance" score. With FAISS, If your question is a close match (a distance score of 0.5 or less), the chatbot instantly pulls the exact pre-vetted answer. If your question doesn't closely match any pre-made pairs (a score above 0.5), a fallback safety net is triggered. The system automatically switches over to search through a secondary database containing raw text chunks from 17 scraped Bury College web pages, ensuring it can still give you an accurate, reliable answer for unexpected questions. If I had used Chroma, one of the differences is in the scores. For Chroma I would look at a score greater than 0.5.

Planning

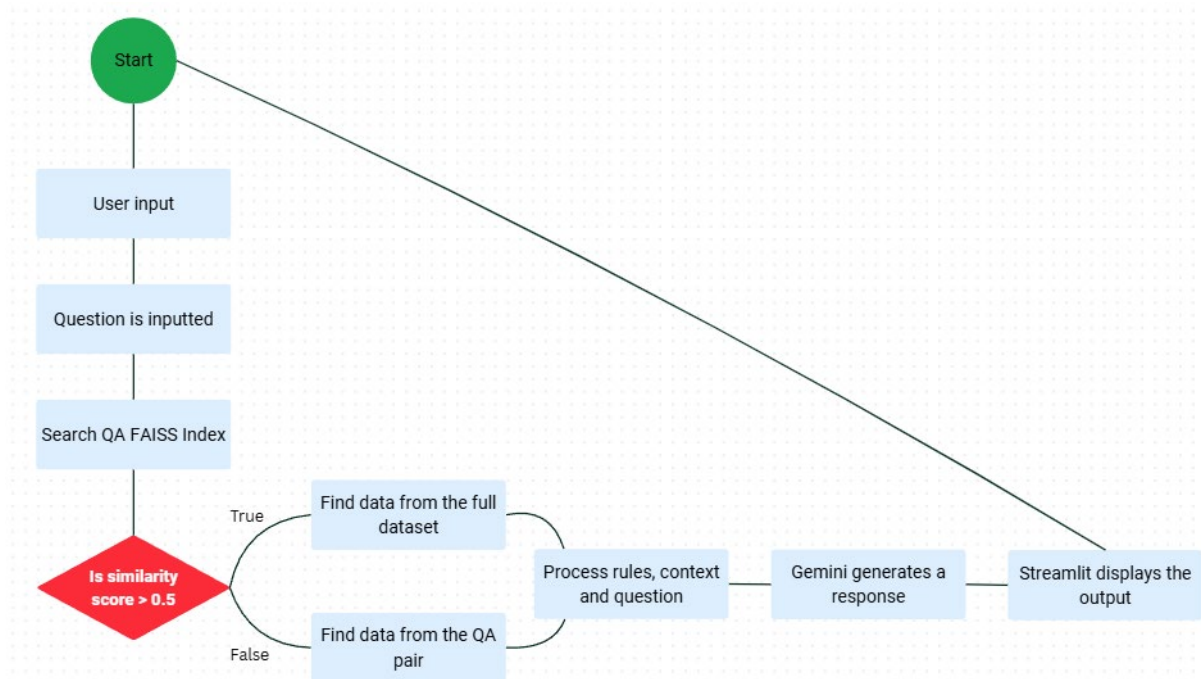
I'll have to think about the libraries I'll need to use and install these. I should also consider creating a separate virtual environment with my Jupyter Notebook to prevent dependency drift and global namespace pollution. I'll investigate BeautifulSoup for text analyses in webpage format, Chroma or FAISS for the creation of the vector database (the former builds the database on my SSD as it is building it where as FAISS uses my RAM). Langchain will be useful to connect the AI model to the data I hold which include the data chunking, component chaining, memory management and autonomous actions so the model can decide on which tool to use and when if faced with a complex problem. Pandas can help with structuring the data allowing it to be cleaned efficiently. Streamlit will be used for the user interface, so the user is not faced with a Python notebook. python-dotenv can keep my API key private. Scikit-Learn contains many mathematical functions. We've been asked to use cosine similarity in the construction of the vector database so this will allow the program to manually calculate exactly how mathematically close two text vectors are to one another.

I'm aiming to use Gemini-3.5-flash because it can comfortably translate my 17 webpages if the number of tokens comes to around 12000 tokens. It also allows for a 1 million token context window which is more than enough for this chatbot exercise. I'll use the embedding system of Gemini-embedding-001 with time.sleep to allow the text chunks without breaching the 15 requests per minute ceiling I would have with my free account.

This is the order in which the program should flow:

- Scrape the data and store this as a JSON file in a location called `data/raw/scraped_pages.json`
- Obtain the API key from Google
- Create the Q+A document from the .json file and save this at a location: `data/processed/qa_dataset.csv`
- Chunk the data and save this: `data/processed/chroma` or `faiss`
- Create the vector database and set the threshold at .5. Anything lower means a match has been found on the Q+A document and we return the Q+A match. Anything higher and we fallback to the full document vector search
- The context is uploaded to the chatbot so it understands what role it should play and the format of the results it should output to the type of audience who might be using it
- I should ensure the FAQ database is saved on my SSD regardless of whether Chroma or FAISS is used
- A final check to ensure the prompts to the chatbot works
- Streamlit file is created to provide the user interface in which I can add a welcome message, dialogue box and other statistics which could be useful. I'll also include citations as to where the output was retrieved.

Diagram of user journey



Query Ingestion & Embedding: The process begins in the Streamlit user interface where the user submits a question. This query is transformed into a vector embedding using the same embedding model utilised during the data ingestion phase so the question can be matched to the data held in the vector database

Synthetic Q/A Search: The embedded user query is first run against the FAISS vector database containing the 204 synthetically generated question-and-answer pairs. The system calculates the highest cosine similarity score among the entries.

If Score ≤ 0.5 : The system a match. The lower the number, the bigger the match in the pre-generated Q/A dataset. The corresponding answer and its source webpage are pulled.

If Score > 0.5 (Fallback Triggered): The system determines that the specific question was not covered in the dataset that was created in my qa_dataset.csv file. It shifts to the fallback layer, running a vector search across the broader database containing chunks of the 17 raw scraped Bury College web pages to gather semantic context.

Augmentation & Generation: The retrieved context (whether from the Q/A tier or the raw document chunks) is combined with a structured system prompt and the user's original query. The correct data is then transformed into a language the user can understand, and using the rules I specified in my context, and outputs the answer.

Model Building

Please see the code on github for this:

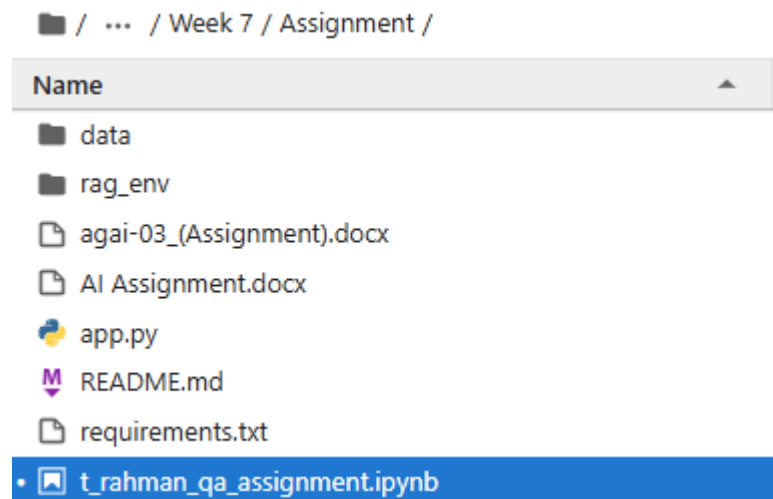
Presentation

Below are screenshots from the folder structure, output of the Jupyter Notebook, the qa_dataset.csv file which was generated and Streamlit in action.

Jupyter Outputs

The folder structure used to store my files is below:

This holds my notes from week 7 but also my assignment folder.



In the assignment folder are my data folder which store the raw and processed data from the data scrape and the vector database which is created later using FAISS. The rag_env folder has my python environment set up so I'm more sure that conflicts will not arise between what I'm using for the assignment and my other Python work. Lastly, my .env file is in this location as well.

Folder structure: / ... / Week 7 / Assignment /

Name	Last Modified
data	2 days ago
rag_env	2 days ago
agai-03_(Assignment).docx	2 days ago
AI Assignment.docx	seconds ago
app.py	24 minutes ago
t_rahman_qa_assignment.ipynb	4 minutes ago

Inside the data folder are the processed and raw folders

📁 / ... / Assignment / data /

Name	Last Modified
📁 processed	7 minutes ago
📁 raw	a day ago

In the processed folder you'll see Chroma_db and Chroma_db_clean. I tried this on the first attempt but there was a conflict between OneDrive constantly updating my SSD and chroma trying to make the vector database. I switched to FAISS as this used my RAM. As I had the storage room to do this, it was fine but I appreciate that if I were to use a much bigger dataset for vectorisation, I would not be able to rely on the 16GB of memory. Using my 4Tb SSD might be the only way forward.

📁 / ... / data / processed /

Name	Last Modified
📁 chroma_db	a day ago
📁 chroma_db_clean	19 hours ago
📁 faiss_db	19 hours ago
📁 faiss_qa_db	19 hours ago
📄 qa_dataset.csv	21 minutes ago

In the raw folder below, this contains the scraped_pages.json file. This will store the dictionary containing the URLs and their raw content. This raw data will be used by the LLM to come up with the Q+A document that will be placed into the processed folder.

📁 / ... / data / raw /

Name	Last Modified
📄 scraped_pages.json	34 minutes ago

Jupyter Notebook

Part 1 – Confirmation that extra libraires which needed to be installed, did so successfully

```
C:\Users\joey\AppData\Local\Temp\ipykernel_25356\1160094315.py:6: DeprecationWarning: 'langchain-community' is being sunset and is no longer actively maintained. See https://github.com/langchain-ai/langchain-community/issues/674 for details and migration guidance toward standalone integration packages.
import langchain_community
All libraries are installed and are accessible in this notebook.
```

Part 2a – Evidence that the data scrape worked on all pages

```
Starting scraper...
Scraping: https://burycollege.ac.uk/16-18
Scraping: https://burycollege.ac.uk/apprenticeships
Scraping: https://burycollege.ac.uk/adults
Scraping: https://burycollege.ac.uk/university-centre
Scraping: https://burycollege.ac.uk/employers
Scraping: https://burycollege.ac.uk/contact-us/contact-information
Scraping: https://burycollege.ac.uk/about-us/workforus
Scraping: https://burycollege.ac.uk/taking-teaching-further
Scraping: https://burycollege.ac.uk/greater-manchester-institute-of-technology
Scraping: https://burycollege.ac.uk/about-us/governance
Scraping: https://burycollege.ac.uk/about-us/our-vision
Scraping: https://burycollege.ac.uk/about-us/ofsted
Scraping: https://burycollege.ac.uk/about-us/term-dates
Scraping: https://burycollege.ac.uk/about-us/sustainability
Scraping: https://burycollege.ac.uk/support-for-schools
Scraping: https://burycollege.ac.uk/events
Scraping: https://burycollege.ac.uk/news
```

Success! Successfully scraped 17 pages and saved to data/raw/scraped_pages.json

Part 2b – Evidience tha the data scrape work as the output of the first three scrapes is present

Total pages found in JSON: 17

```
--- Page 1 ---
URL: https://burycollege.ac.uk/16-18
Content Character Length: 3899
Snippet: School Leavers (16-18) - Bury College Skip to main content Skip to search Skip to navigation Open action menu Campus map Contact us Course guides Appl...

--- Page 2 ---
URL: https://burycollege.ac.uk/apprenticeships
Content Character Length: 6643
Snippet: Apprenticeships - Bury College Skip to main content Skip to search Skip to navigation Open action menu Campus map Contact us Course guides Apply now A...

--- Page 3 ---
URL: https://burycollege.ac.uk/adults
Content Character Length: 4884
Snippet: Adult courses - Bury College Skip to main content Skip to search Skip to navigation Open action menu Campus map Contact us Course guides Apply now Adu...
```

Part 3 – Confirmation that my API key works

```
# Ensure Googles API key is found and works
from dotenv import load_dotenv # Enables me to secure my API key when called

# Run load_dotenv into the environment so my program can pick up my API key
load_dotenv()

# Retrieve the key to verify it loaded correctly
google_api_key = os.getenv("GOOGLE_API_KEY")

if google_api_key:
    print("API Key set successfully!")
else:
    print("Error with the API key")
```

API Key set successfully!

Part 4 – Confirmation of the Q+A document using the 17 webpages

```
Starting Synthetic Q/A Generation with production model...
[1/17] Generating Q/As for: https://burycollge.ac.uk/16-18
[2/17] Generating Q/As for: https://burycollge.ac.uk/apprenticeships
[3/17] Generating Q/As for: https://burycollge.ac.uk/adults
[4/17] Generating Q/As for: https://burycollge.ac.uk/university-centre
[5/17] Generating Q/As for: https://burycollge.ac.uk/employers
[6/17] Generating Q/As for: https://burycollge.ac.uk/contact-us/contact-information
[7/17] Generating Q/As for: https://burycollge.ac.uk/about-us/workforus
[8/17] Generating Q/As for: https://burycollge.ac.uk/taking-teaching-further
[9/17] Generating Q/As for: https://burycollge.ac.uk/greater-manchester-institute-of-technology
[10/17] Generating Q/As for: https://burycollge.ac.uk/about-us/governance
[11/17] Generating Q/As for: https://burycollge.ac.uk/about-us/our-vision
[12/17] Generating Q/As for: https://burycollge.ac.uk/about-us/ofsted
[13/17] Generating Q/As for: https://burycollge.ac.uk/about-us/term-dates
[14/17] Generating Q/As for: https://burycollge.ac.uk/about-us/sustainability
[15/17] Generating Q/As for: https://burycollge.ac.uk/support-for-schools
[16/17] Generating Q/As for: https://burycollge.ac.uk/events
[17/17] Generating Q/As for: https://burycollge.ac.uk/news
```

Success! Generated 204 Q/A pairs and saved to data/processed/qa_dataset.csv

Part 5 – Evidence that the vector database is built. Batches used so to not exhaust my quota or overload the server

```
Connecting to Google Embeddings API...
Loading full document vector database...
Loading synthetic Q/A dataset from: data/processed/qa_dataset.csv
Building semantic search index for 204 Q/A pairs safely in batches...
-> Initializing Q/A batch 1...
-> Processing Q/A batch 2/41...
-> Processing Q/A batch 3/41...
-> Processing Q/A batch 4/41...
-> Processing Q/A batch 5/41...
-> Processing Q/A batch 6/41...
-> Processing Q/A batch 7/41...
-> Processing Q/A batch 8/41...
-> Processing Q/A batch 9/41...
-> Processing Q/A batch 10/41...
-> Processing Q/A batch 11/41...
-> Processing Q/A batch 12/41...
-> Processing Q/A batch 13/41...
-> Processing Q/A batch 14/41...
-> Processing Q/A batch 15/41...
-> Processing Q/A batch 16/41...
-> Processing Q/A batch 17/41...
-> Processing Q/A batch 18/41...
-> Processing Q/A batch 19/41...
-> Processing Q/A batch 20/41...
-> Processing Q/A batch 21/41...
-> Processing Q/A batch 22/41...
-> Processing Q/A batch 23/41...
-> Processing Q/A batch 24/41...
-> Processing Q/A batch 25/41...
-> Processing Q/A batch 26/41...
-> Processing Q/A batch 27/41...
-> Processing Q/A batch 28/41...
-> Processing Q/A batch 29/41...
-> Processing Q/A batch 30/41...
-> Processing Q/A batch 31/41...
-> Processing Q/A batch 32/41...
-> Processing Q/A batch 33/41...
-> Processing Q/A batch 34/41...
-> Processing Q/A batch 35/41...
-> Processing Q/A batch 36/41...
-> Processing Q/A batch 37/41...
-> Processing Q/A batch 38/41...
-> Processing Q/A batch 39/41...
-> Processing Q/A batch 40/41...
-> Processing Q/A batch 41/41...
```

Both retrieval layers are loaded, aligned, and completely ready!

Part 6 – This was verifying that the context window was built and would know how to format the response to the questions asked.

Initializing Gemini LLM...

Part 7 – Evidence that my vector database is stored locally on my SSD

```
qa_vector_store.save_local("data/processed/faiss_qa_db")
print("QA Vector Store saved successfully!")
```

QA Vector Store saved successfully!

Part 8 – Evidence of testing

Question 1 – This was a question which was not on my Q+A and therefore showed a weak confidence level at 0.62 so the output was retrieved from the full document

```
bury_college_chatbot("What courses are available?")

[User Query]: 'What courses are available?'
-> [Stage 1 - Q/A Match] Nearest question found with distance score: 0.6146
[Confidence low or no exact match. Falling back to full document chunk search...]
[ Gemini is generating your answer based on retrieved facts... ]

=====
BURY COLLEGE ASSISTANT RESPONSE:
=====
Bury College offers a wide range of courses across various subject areas. Based on our current listings, here are the courses available:

**Health and Social Care**
* Health and Social Care (Level 2 and Level 3)

**Catering and Hospitality**
* Professional Cookery (Level 2 and Level 3)
* Hospitality and Catering (Level 2)

**Construction and the Built Environment**
* Bricklaying (Level 2)
* Carpentry and Joinery (Level 2 and Level 3)
* Electrical Installation (Level 2 and Level 3)
* Plastering (Level 2)
* Plumbing (Level 2 and Level 3)

**English and Maths**
* GCSE English (Level 2)
* Functional Skills English (Level 2)
* GCSE Maths (Level 2)
* Functional Skills Maths (Level 2)

**Special Educational Needs and Disabilities (SEND)**
* Prospects Programme - featuring vocational pathways in Business, Catering, Childcare, ICT, and Sport (Level 2)
* Prospects Construction Programme (Level 2)

**Adult and Distance Learning**
* Free online distance learning Level 2 courses (completed at home or in the workplace)

If you have any questions about these courses or the application process, please feel free to contact our Admissions team at admissions@burycollege.ac.uk or call 0161 280 8200.
=====
Source Type: Full Web Page Scrapes
Verified Source URL(s): https://burycollege.ac.uk/taking-teaching-further, https://burycollege.ac.uk/adults
=====
```

Question 2 – The next question was taken, almost word for word, from the qa_dataset.csv file on question 4. As expected, the score is 0.35 which is much higher in confidence so it could bypass the full document and refer to the answer in the qa_dataset.csv file.

```
bury_college_chatbot("Can I earn money while studying an apprenticeship?")

[User Query]: 'Can I earn money while studying an apprenticeship?'
-> [Stage 1 - Q/A Match] Nearest question found with distance score: 0.3493
[High-confidence match found in Q/A dataset! Bypassing document search.]
[ Gemini is generating your answer based on retrieved facts... ]

=====
BURY COLLEGE ASSISTANT RESPONSE:
=====
Yes, you can! An apprenticeship at Bury College is a job where you will have the opportunity to learn new skills, gain experience, and get paid while working towards your chosen career.
=====
Source Type: Curated Q/A Pair
Verified Source URL(s): https://burycollege.ac.uk/16-18
=====
```

Question 3 – The final question was taken word for word from the qa_dataset.csv file. Confidence was even better at 0.26 and again, the response was as is found in the answer column on the qa_dataset.csv file.

```
bury_college_chatbot("What are the different study pathways available for 16-18 year old school leavers at Bury college?")

[User Query]: 'What are the different study pathways available for 16-18 year old school leavers at Bury college?'
-> [Stage 1 - Q/A Match] Nearest question found with distance score: 0.2561
High-confidence match found in Q/A dataset! Bypassing document search.
Gemini is generating your answer based on retrieved facts...

=====
BURY COLLEGE ASSISTANT RESPONSE:
=====
Welcome to Bury College!

For 16-18 year old school leavers, we offer four distinct study pathways:

* **A Levels**
* **Vocational Programmes**
* **T Levels**
* **Apprenticeships**
=====
Source Type: Curated Q/A Pair
Verified Source URL(s): https://burycollege.ac.uk/16-18
=====
```

Based on these three tests, I would be confident in putting this system live

Part 9 - Streamlit now installed

```
!rag_env\Scripts\pip install streamlit
```

Qa_dataset.csv sample

The samples below show the actual Q+A pairs that have been made from 1 to 204. These questions help me to test the chatbots accuracy in Jupyter before I go ahead and build the frontend using Streamlit.

t_rahman_qa_assignment.ipynb qa_dataset.csv			
Delimiter: ,			
	question	answer	source_page
1	What are the different study pathways available for 16-18 year old school leavers at Bury College?	Bury College offers fo...	https://burycollege.ac.uk/16-18
2	Aside from my main qualification, what else will my study programme at Bury College include?	Your study programm...	https://burycollege.ac.uk/16-18
3	How do T Levels work at Bury College, and how do they differ from standard A-levels?	T Levels are high-qua...	https://burycollege.ac.uk/16-18
4	Can I earn money while studying an apprenticeship at Bury College?	Yes, an apprenticeshi...	https://burycollege.ac.uk/16-18
5	I am an adult looking to return to education. Does Bury College host any events where I can find o...	Yes, Bury College is h...	https://burycollege.ac.uk/16-18
6	Where can I find the course guide for entry in September 2026?	You can download the...	https://burycollege.ac.uk/16-18
7	As a local business, can we partner with Bury College to hire apprentices?	Yes, Bury College par...	https://burycollege.ac.uk/16-18
8	How do Bury College students contribute to the local community?	Bury College students...	https://burycollege.ac.uk/16-18
9	What can I expect from a Vocational study programme at Bury College?	Vocational study prog...	https://burycollege.ac.uk/16-18
10	What trades were represented in the GM Skills Construction Competition hosted by Bury College i...	The GM Skills Constr...	https://burycollege.ac.uk/16-18
11	Does Bury College support Foundation Studies learners in regional competitions?	Yes, Bury College hos...	https://burycollege.ac.uk/16-18
12	Does Bury College offer options for higher education or teacher training?	Yes, Bury College Uni...	https://burycollege.ac.uk/16-18
13	I am currently 16 and about to leave school. Am I eligible to apply for an apprenticeship at Bury Co...	Yes, apprenticeships ...	https://burycollege.ac.uk/apprenticeships
14	I am an adult looking to change careers. Can I still do an apprenticeship, or are they only for youn...	Yes, apprenticeships ...	https://burycollege.ac.uk/apprenticeships
15	How much of an apprentice's week is spent on off-the-job training at Bury College?	Apprentices spend a ...	https://burycollege.ac.uk/apprenticeships
16	What is the starting pay rate for an apprentice at Bury College?	You will earn a minim...	https://burycollege.ac.uk/apprenticeships
17	We are a local business looking to hire an apprentice. How does Bury College support employers ...	Bury College's Appre...	https://burycollege.ac.uk/apprenticeships
18	What qualifications do I need to start a Level 3 apprenticeship at Bury College?	To start a Level 3 app...	https://burycollege.ac.uk/apprenticeships
19	What educational qualification will I receive if I complete a Level 2 apprenticeship?	Completing a Level 2 ...	https://burycollege.ac.uk/apprenticeships
20	How are apprentices assessed throughout their programme?	Assessment takes pla...	https://burycollege.ac.uk/apprenticeships

t_rahman_qa_assignment.ipynb x qa_dataset.csv x +			
Delimiter: , v			
	question	answer	source_page
181	I am an adult looking to return to education. Is there an event where I can explore my options for c...	Yes, Bury College is h...	https://burycollege.ac.uk/events
182	How does Bury College support local care leavers through its student projects?	Bury College art stud...	https://burycollege.ac.uk/events
183	Do current apprentices at Bury College get to hear from former students who have successfully co...	Yes, former Bury Coll...	https://burycollege.ac.uk/events
184	I want to become a teacher. Can Bury College University Centre help me achieve this career goal?	Yes, Bury College Uni...	https://burycollege.ac.uk/events
185	As a local childcare employer, how can Bury College help me recruit and train new staff?	Bury College partners...	https://burycollege.ac.uk/events
186	What kind of activities do Foundation Studies students participate in during regional competitions a...	During the GM Skills ...	https://burycollege.ac.uk/events
187	Which trade areas were represented in the GM Skills Construction Competition hosted by Bury Co...	The competition featu...	https://burycollege.ac.uk/events
188	How many colleges took part in the GM Skills Construction Competition held at Bury College in Ma...	Students from nine dif...	https://burycollege.ac.uk/events
189	When is the upcoming Adult Learning Open Event being held?	The Adult Learning O...	https://burycollege.ac.uk/events
190	What was the purpose of the GM Skills Foundation Studies Competition hosted at the college on A...	The event was design...	https://burycollege.ac.uk/events
191	How long has the Mulberry Bush Nursery Group been partnering with Bury College for childcare a...	According to the colle...	https://burycollege.ac.uk/events
192	Where did the Bury College art students paint their murals to help the local community?	The art students creat...	https://burycollege.ac.uk/events
193	My daughter is interested in studying Art and Design. Do Bury College students get opportunities t...	Yes, Bury College art ...	https://burycollege.ac.uk/news
194	Do you ever invite former apprentices back to the college to speak with current students?	Yes, Bury College reg...	https://burycollege.ac.uk/news
195	I am an adult learner hoping to transition into teaching. Does Bury College University Centre offer ...	Yes, Bury College Uni...	https://burycollege.ac.uk/news
196	As a local business owner, I want to know if Bury College has experience partnering with employer...	Bury College has a lo...	https://burycollege.ac.uk/news
197	What kind of activities and events do Foundation Studies students participate in at Bury College?	Foundation Studies st...	https://burycollege.ac.uk/news
198	I am interested in a construction trade. What specific areas were featured in the GM Skills Constr...	The GM Skills Constr...	https://burycollege.ac.uk/news
199	As a college governor, I'd like to know how many regional colleges participated in the GM Skills Co...	Bury College hosted s...	https://burycollege.ac.uk/news
200	I am looking to return to study as an adult. Is there an upcoming open event where I can speak to t...	Yes, Bury College is h...	https://burycollege.ac.uk/news
201	Does Bury College offer apprenticeships in professional services like finance or accounting?	Yes, Bury College off...	https://burycollege.ac.uk/news
202	I want to start a career in the healthcare sector. Does the college offer specific pathways or suppor...	Yes, Bury College off...	https://burycollege.ac.uk/news
203	What creative and performing arts subject options are available for study at Bury College?	Bury College offers se...	https://burycollege.ac.uk/news
204	Are there any apprenticeship options available in the construction trades at Bury College?	Yes, Bury College off...	https://burycollege.ac.uk/news

Streamlit Examples

Running Streamlit from the command prompt:

- `cd "OneDrive\Documents\Home\Training\AI\Codecademy\Week 7\Assignment"`
- `rag_env\Scripts\streamlit run app.py`

```

Command Prompt - rag_env\Scripts\streamlit run app.py
Microsoft Windows [Version 10.0.19045.7291]
(c) Microsoft Corporation. All rights reserved.

C:\Users\joey>cd "OneDrive\Documents\Home\Training\AI\Codecademy\Week 7\Assignment"

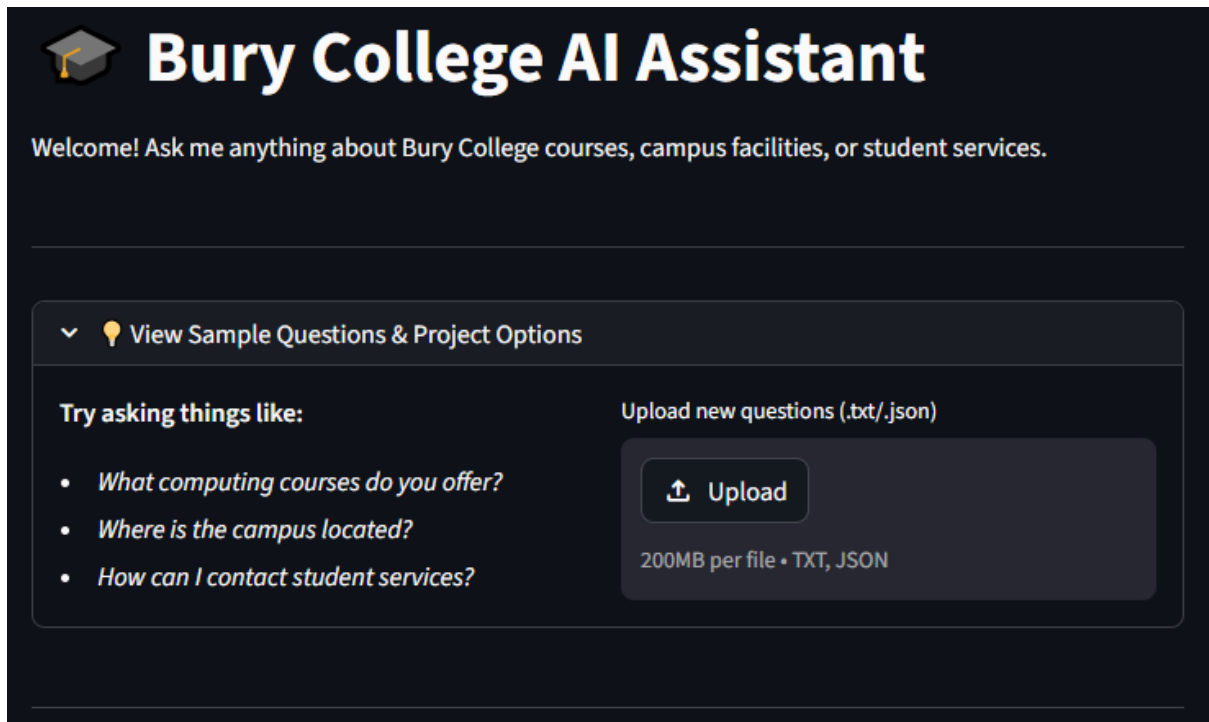
C:\Users\joey\OneDrive\Documents\Home\Training\AI\Codecademy\Week 7\Assignment>rag_env\Scripts\streamlit run app.py
2026-05-28 14:34:29.388 Uvicorn server started on 0.0.0.0:8501

You can now view your Streamlit app in your browser.
Local URL: http://localhost:8501
Network URL: http://192.168.1.246:8501

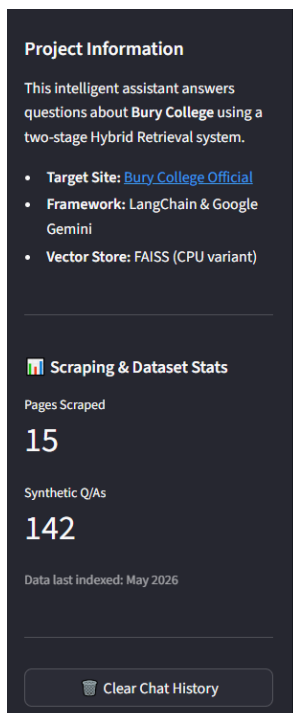
C:\Users\joey\OneDrive\Documents\Home\Training\AI\Codecademy\Week 7\Assignment\app.py:10: DeprecationWarning: `langchain-community` is being sunset and is no longer actively maintained. See https://github.com/langchain-ai/langchain-community/issues/674 for details and migration guidance toward standalone integration packages.
  from langchain_community.vectorstores import FAISS
  
```


Streamlit in action

When the menu is clicked at the top, you can see sample questions and also upload new questions (but the uploaded document does not update the RAG).



Sidebar on the left



Question from the qa_dataset.csv file



What are the different study pathways available for 16-18 year old school leavers at Bury college?



Welcome to Bury College!

For 16-18 year old school leavers, we offer four distinct study pathways:

- A Levels
- Vocational Programmes
- T Levels
- Apprenticeships

If you have any questions about these pathways or need help choosing the right one for you, please feel free to ask!

Source Type: Curated Q/A Pair | Verified URL(s): <https://burycollege.ac.uk/16-18>

Question which is not on the qa_dataset.csv file



What a-levels do the college run?



While the provided information highlights that the Bury College A Level Centre offers engaging, interesting, and personalised study programmes, it does not list the specific A-level subjects currently available.

To find the full list of A-level courses we offer, we recommend contacting the college directly or visiting the A Levels section on our website.

Source Type: Full Web Page Scrapes | Verified URL(s): <https://burycollege.ac.uk/greater-manchester-institute-of-technology>, <https://burycollege.ac.uk/16-18>

Remembers chat history



My name is Timur and I'm 18. Please reference my name in future responses.



Hello Timur! It is a pleasure to meet you.

As you are 18, you fall within our School Leavers (16-18) age group. At Bury College, we offer several pathways tailored to your individual talents and career ambitions, including A Levels, Vocational programmes, T Levels, and Apprenticeships. In addition to your qualifications, your study programme here would include career-enhancing work placements, exciting enrichment opportunities, impartial careers advice, and access to our wellbeing and student support services.

How can I help you explore your options today, Timur?

i Source Type: Full Web Page Scrapes | Verified URL(s): <https://burycollege.ac.uk/16-18>, <https://burycollege.ac.uk/support-for-schools>



What is it like to work here. Remember to use my name.



Hello Timur!

At Bury College, we believe our staff are our number one asset, and we are dedicated to attracting, developing, and retaining excellent people who support each other to deliver our mission of serving our communities through education and training.

Operationalise

How to use this

As this is not deployed to a live server, I had to inform the PC as to where my .ipynb and other assignments files were located in the command prompt (below):

```
cd "OneDrive\Documents\Home\Training\AI\Codecademy\Week 7\Assignment"
```

Then to run the application, I wrote into the command prompt the following as this is where the app.py file is located on my system. You will note that I used relative paths for flexibility:

```
rag_env\Scripts\streamlit run app.py
```

Issues when building

Using an API from Gemini was great in terms of being free to use but it does have strict quotas, on the first and second round of testing my code, I would hit these quotas. I could switch models to extend this time but there are only so many models you can use before I would have to wait until the next day for the quota to refresh. If I were to do this project for real, I would need to know from the stakeholders, how often their website is visited and that would help me to understand the quotas they would need.

Chroma was used to begin with when building the vector database but there was a conflict with OneDrive. So as OneDrive was constantly synchronising my data in the cloud, it was locking out Chroma from making continual appending actions to the vector database. I switched to FAISS as this built the database in RAM. This is great for small databases but not for larger ones. In the future, to get around this, I could disable the sync or work in a folder which is outside of the OneDrive area.

Lastly, while I could get Streamlit to include a facility to upload a .txt or .csv file, I could not get it to update the qa_dataset.csv file.

Future proofing the chatbot

Only 17 pages were uploaded. The college has 54 webpages, 5 social media links and various portals and documents attached to the webpages. If this chatbot were to be more useful, I would need to upload all the pages, documents and also have the content on the social media channels scanned regularly.

Finally, a chatbot could be used for other purposes such as for employees to check human resources documents, policy documents and curriculum.

Success Criteria Revisited

I was asked to do the following elements which work are in green and those that partly worked are in orange. Nothing in red as there wasn't anything that was asked for which is missing or completely broken:

- Use Python (Anaconda JupyterLab)
- Install the relevant libraries and run them into memory
- Data scraping of a website (<https://burycollege.ac.uk/>) and a minimum of 10 webpages
- Clean the data so I'm only left with the text and then save this as .txt or .json file
- Use of a LLM to generate high quality question-answer pairs from the scraped content with a target of between 100 – 200 question and answer pairs
- The Q+A document can be saved as a .csv or .json file
- Create the vector database by chunking the scraped documents
- Create embeddings and then store this in a vector database which will be done in either Chroma or FAISS
- The chatbot itself needs to search the QA database. If the confidence is a low to no good match, then the chatbot should fall back on to the vector database which stores all of the web scrapped information
- The LLM will then piece together the words it needs for the solutions from the QA database or vector database and return this information back to the user
- There needs to be chat history
- The user interface should use Streamlit and feature:
 - chat interface
 - source citation (based on the Q+A document)
 - clear chat button
 - a side bar with project information
 - website link and statistics
 - option to upload new questions or view a sample of the Q+As – you can upload files but these are not utilised by the chatbot but this placeholder allows me to build in backend functionality at a later date.